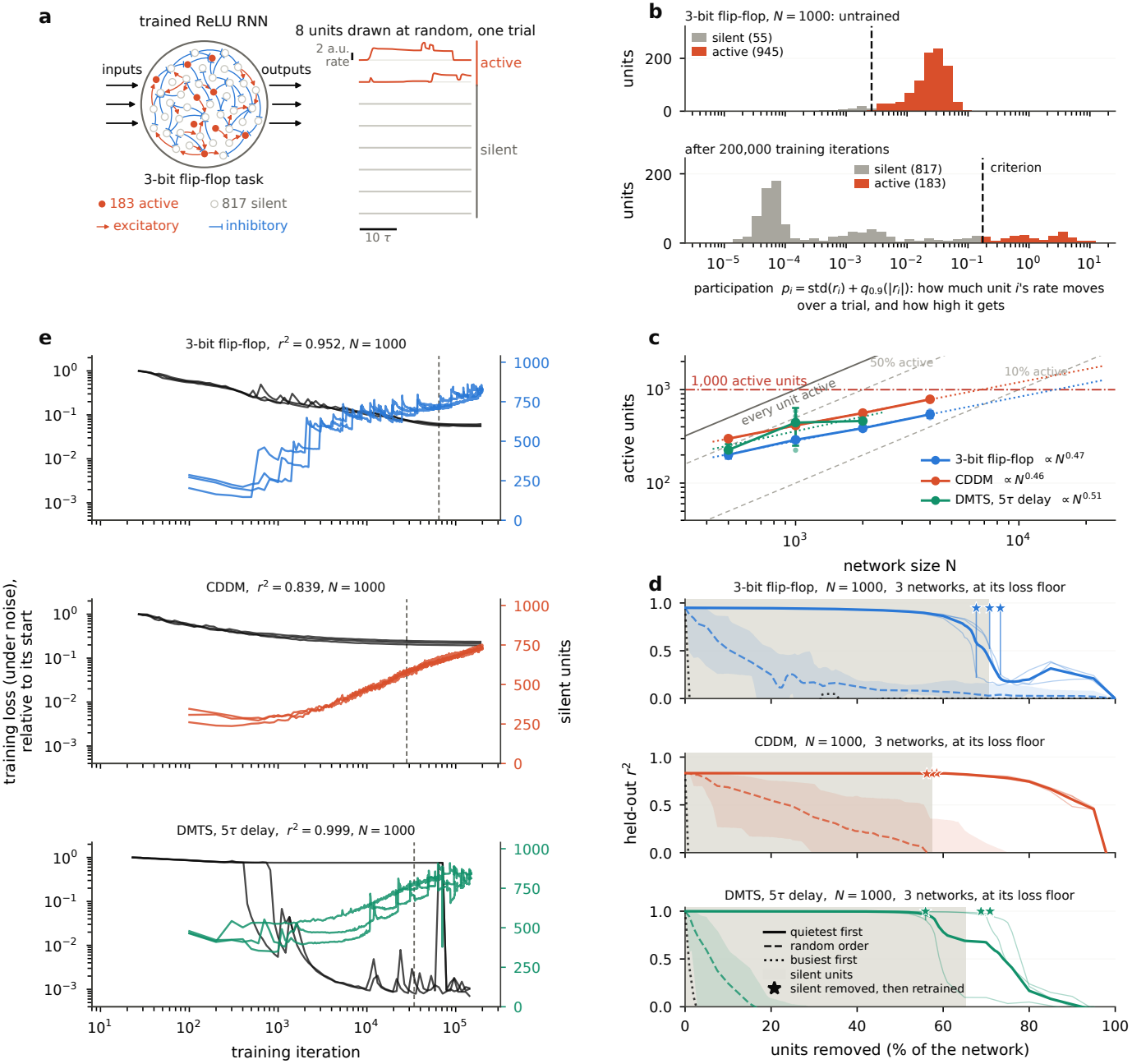

a
trained ReLU RNN
inputs
outputs
3-bit flip-flop task
183 active
817 silent
excitatory
inhibitory
8 units drawn at random, one trial
2 a.u.
rate
active
silent
10 τ
b
3-bit flip-flop, N = 1000: untrained
silent (55)
active (945)
units
after 200,000 training iterations
silent (817)
active (183)
criterion
participation $p_i = \mathrm{std}(r_i) + q_{0.9}(|r_i|)$: how much unit i's rate moves over a trial, and how high it gets
e
3-bit flip-flop, $r^2 = 0.952$, $N = 1000$
CDDM, $r^2 = 0.839$, $N = 1000$
DMTS, 5τ delay, $r^2 = 0.999$, $N = 1000$
training loss (under noise), relative to its start
silent units
training iteration
c
1,000 active units
every unit active
50% active
10% active
active units
3-bit flip-flop $\propto N^{0.47}$
CDDM $\propto N^{0.46}$
DMTS, 5τ delay $\propto N^{0.51}$
network size N
d
3-bit flip-flop, N = 1000, 3 networks, at its loss floor
CDDM, N = 1000, 3 networks, at its loss floor
DMTS, 5τ delay, N = 1000, 3 networks, at its loss floor
held-out r^2
quietest first
random order
busiest first
silent units
silent removed, then retrained
units removed (% of the network)